Site Overview

The 430-460 High Street, Medford, Massachusetts site is a former industrial site that currently has multiple businesses running on the property. This site was contaminated with chlorinated solvents in shallow groundwater. The most prominent one of these is trichloroethene (TCE), which has the highest levels of contamination shown below (Figure 1).

Trichloroethene is a chemical used to make refrigerants and is not naturally occurring. It is extremely volatile and is a colorless, liquid chemical that easily sorbs into soil and groundwater. This makes it an extremely hazardous chemical and must be removed from the groundwater.

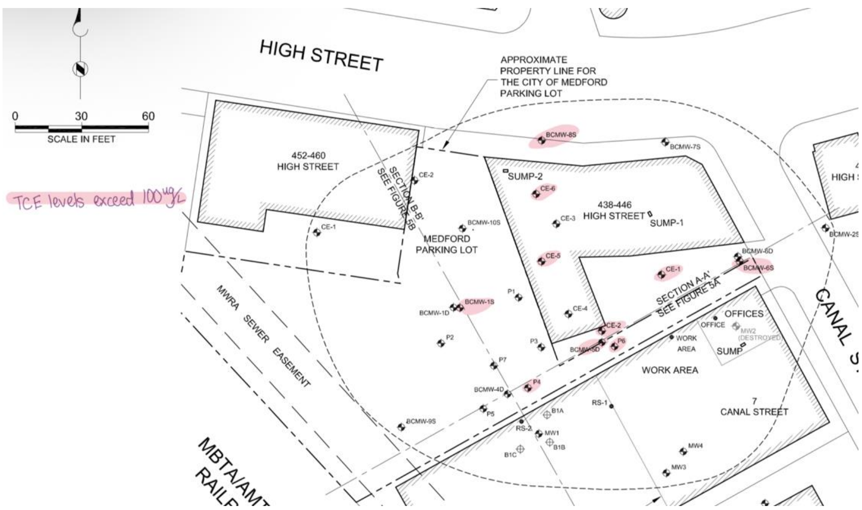

Figure 1. TCE levels on High St Site

The objective of this project is to remove as much TCE as possible from the groundwater until it reaches legal limits. The legal limit for TCE in groundwater is 5 ug/L (MassDEP).

Site Conceptual Model

The geology of the site is that there is a layer of sandy aquifer around 0.25-5ft bgs, followed by a layer of silt and sand from 5-15ft below ground surface. The water table is at about 11.5 ft bgs. The groundwater seems to flow in the northeast direction when using contamination analysis. The TCE concentrations detected in the ground go up to 29,000 mg TCE/kg of soil.

Selected Technology

The selected technology for this project is a pump and treat system. The pump and treat will be installed with multiple extraction wells in the most highly concentrated area of TCE levels on the site.

Conceptual Design

The chosen method is to install 4 extraction wells within a 70x70ft area with a 35’ radius of influence (see Figure 2). These wells will target the areas with the highest concentrations of TCE. There will be a tank placed in the center in order to power the pumping process. We will remove the contaminated water using an air stripper connected to the AOU. Then, treated water will be reinjected upgradient to create a hydraulic barrier. We will use 1 ton of GAC as an adsorbent in order to ensure no contamination is left.

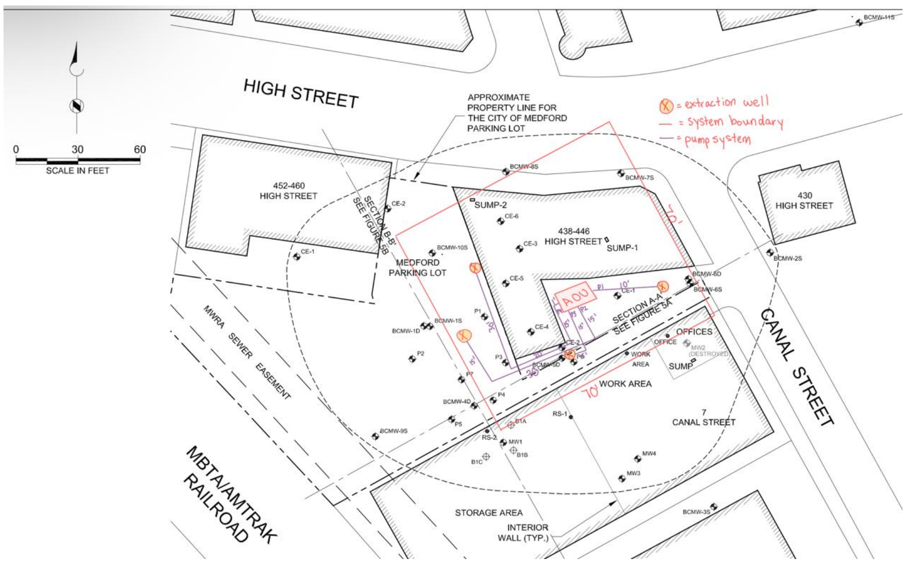

Figure 2. Trench Location of Pump and Treat System

The system is designed to operate for about 20 years. Pump and treat systems are proven technology that are designed to treat areas with high VOC concentrations, which is perfect for this site.

Cost Estimate

Typically, the average annual cost of a pump and treat system is $570,000. Using cost values from Inogen Alliance, Fed Remediation, and US EPA, we were able to estimate the necessary costs for this system (Figure 3).

The labor costs were estimated at $50,000 a year. The costs may be higher due to how much time the system may need to be installed. The value of the proposed pump and treat system is much lower than average. Taking error into consideration (assuming 35%), the total could be as low as $302,547 to $628,368.

Item	Qty	Unit	Unit Cost	Cost
Extraction Wells	4	EA	$ 20,000.00	$ 80,000.00
Piping and Trenching	156	LF	$ 100.00	$ 15,600.00
Air Stripper System	1	LS	$ 123,406.00	$ 123,406.00
GAC Unit	2000	LB	$ 1.87	$ 3,740.00
Control Panel and Electrical	1	LS	$ 7,000.00	$ 7,000.00
Labor Installation	1	LS	$ 50,000.00	$ 50,000.00
Capital Costs	1	YR	$ 75,000.00	$ 75,000.00
O&M Costs	1	YR	$ 50,000.00	$ 50,000.00
Monitoring and Management	15%			$ 60,711.90
Total PV	**$ 465,457.90**			

Figure 3. Cost Estimation

Design Calculations

Pump tube length:

P1 = 10 ft

P2 = 15 ft + 30 ft + 15 ft = 60 ft

P3 = 15 ft + 5 ft = 20 ft

P4 = 1 ft + 15 ft + 20 ft + 30 ft = 66 ft

Carbon Usage: 2000 lb = 9.0718e+11 ug, which will sufficiently exceed the sum of TCE contaminants and more error.

References

MassDEP: https://www.mass.gov/guides/drinking-water-standards-and-guidelines

Inogen: https://www.inogenalliance.com/blog-post/how-much-does-it-cost-site-remediation

Fed Remediation: https://www.frtr.gov/matrix2/section4/Cost_Estimate_2.pdf

US EPA: https://www.epa.gov/sdwa/drinking-water-treatment-technology-unit-cost-models